HEIRSTONE CONSULTING

WORKER HOUSING IN AFRICAN INDUSTRIAL PARKS

Factories need beds

Africa's industrial parks were planned around sheds, power and tax incentives. The workers' beds were left to the market, and the bill comes back to the factory as turnover and transport. A benchmark of 17 zones, the arithmetic of a rental bed and a test for the next generation of parks.



THE REPORT IN SIX NUMBERS

7 dedicated beds per 100 jobs at Ethiopia's Hawassa Industrial Park, where 91% of 33,199 workers came from outside the city

60-120% annual staff turnover reported by Hawassa employers in 2018-19: between 5% and 10% of the workforce leaving every month

120% rent near Bole Lemi as a share of the 750-birr base wage in 2018, when the park had lost 7,840 workers in six months, according to Addis Fortune citing IPDC

US$4,500 cost per bed of a dormitory in Bangladesh that found no tenants (62 to a room) and of an 18,720-bed hostel in Tamil Nadu (six to a room)

5% of Hawassa's 300 hectares would house its whole workforce at the density of Tamil Nadu's Vallam Vadagal hostel

US$68-220 monthly entry wage at which a rent of 20% of pay covers the capital charge on a US$2,100 to 4,500 bed, on assumed finance terms and excluding land and operating costs

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ABOUT THIS REPORT

This report is written for the people who build, fill and finance Africa's industrial parks: zone developers and operators, manufacturers, governments and investment promotion agencies, development finance institutions and housing developers. It draws on public sources current in 2026: park operators and government agencies, statistics offices, development finance institutions, research institutes, company disclosures and the press.

Where we combine published figures into a new one, such as beds per 100 jobs, rent as a share of the wage or the capital charge on a bed, the calculation is labelled Heirstone analysis and its inputs are shown. Local-currency figures carry a US dollar equivalent at the World Bank annual average rate for the year stated.

A separate Heirstone report on export zones covers trade, incentives and investment. This one is about where the workers live.

SUMMARY

The industrial park has a shed for every tenant and a bed for few of their workers

African industrial parks bring tens of thousands of workers to places where they do not live. Their housing is left to private landlords, bus contracts or social housing elsewhere in town, and the cost reaches the factory as turnover, transport and lost output.

1 **Beds are missing from the park model.**

At Hawassa, Ethiopia's flagship park, up to 2,400 dedicated beds (2018) stood against 33,199 workers (2020), 91% of them from outside the city: seven beds per 100 jobs. Benin's GDIZ passed 25,000 jobs in 2026 with no worker housing reported. Egypt's industrial new towns take a city-wide approach: 10th of Ramadan has 194,133 homes built alongside 353,656 factory jobs. No park operator in our benchmark publishes a bed count.

2 **Turnover is where the gap is paid for.**

Hawassa employers reported 5 to 10% of staff leaving every month: 20,000 to 40,000 hires a year on 33,199 jobs, or 8 to 17 per dedicated bed. At Bole Lemi, 7,840 left in six months while nearby rents were about 120% of the base wage (Addis Fortune, citing IPDC).

3 **Shared rooms already cost what a dormitory would.**

A place in a shared room near Hawassa cost 31 to 37% of the 750-birr base wage, close to the 30% the IGC proposed for a low-cost dormitory. A room of one's own cost up to twice the wage. Workers shared three or four to a room, and some walked three hours a day.

4 **A bus is a housing decision made by default.**

Morocco's Atlantic Free Zone moved 25,300 workers with 480 minibuses. Tangier employers paid about EUR 25 a worker a month for buses in 2013, enough to service the capital on a bed costing US$3,400 to 5,100 at assumed rates of 6 to 10%. Buses suit zones beside a city.

5 **Design and operator decide whether workers move in.**

A Bangladesh dormitory and a Tamil Nadu hostel each cost about US$4,500 a bed. The first, 62 beds to a room and let bed by bed by a government directorate, had no tenants after a year. The second, six to a room and run by the anchor employer, opened with 18,720 beds.

6 **Land and wages allow it in more places than assumed.**

At the Tamil Nadu hostel's density, 5% of Hawassa's land would house every worker in the park. On assumed finance terms, a rent of 20% of pay covers a bed's capital charge, before land and operating costs, once entry wages reach US$68 to 220 a month: Egypt, Tunisia and Morocco clear the bar, Benin is close, and Ethiopia and Nigeria need a payer.

7 **The missing piece is a product.**

Kenya's housing levy raised KSh 73.2bn in the year to June 2025, when 1,795 homes were completed. Development finance has gone to home ownership and zone platforms; the one ring-fenced worker dormitory deal we found in Africa is a KfW grant at Hawassa.

WHAT IT MEANS

Zone developers should reserve land for worker beds in the masterplan and report beds per 100 jobs next to jobs and sheds.

Manufacturers should price the bed against turnover and buses, and lease beds for the share of staff who come from far away.

Governments can create the market cheaply: land rules, employer leases and part of existing housing levies.

DFIs and developers have an asset class waiting: employer-leased rental beds at the factory gate, financed in local currency.

17

African zones in the Heirstone Worker Housing Benchmark, in ten countries

1

African park where housing, rent, wages and turnover are all documented: Hawassa

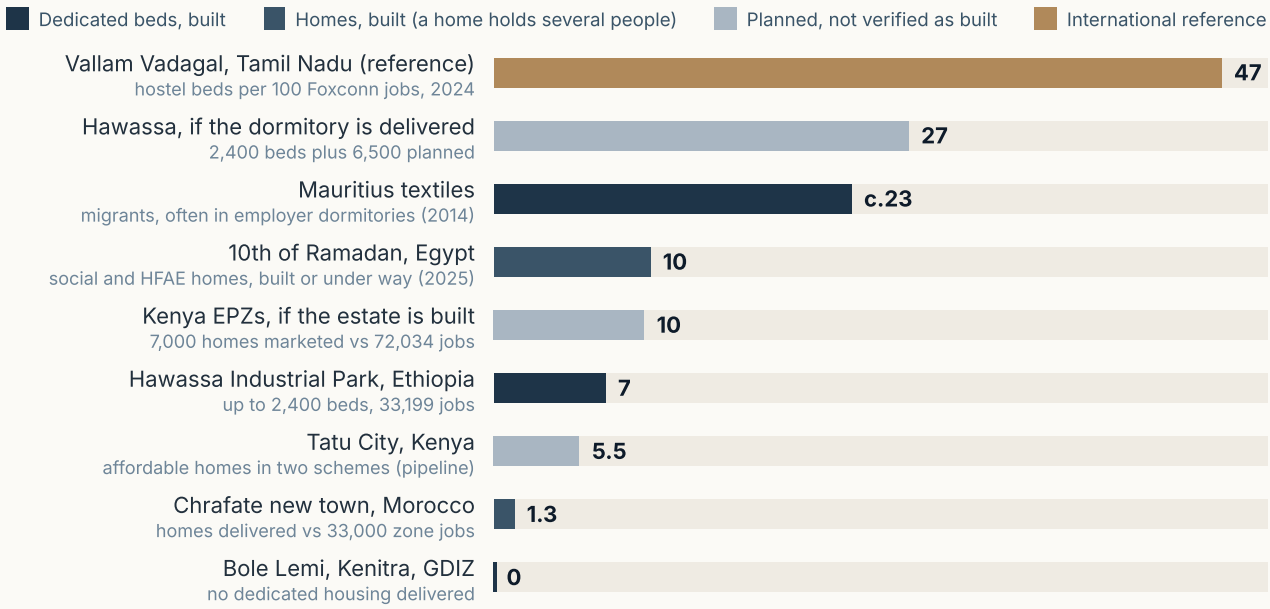
THE MISSING RATIO

Park scorecards count sheds, investors and jobs. The ratio of beds to jobs is missing from all of them.

EXHIBIT 1

Where it can be counted, dedicated housing in African zones runs at single digits per 100 jobs

Dedicated beds, or homes, per 100 jobs at selected zones; latest available year for each input



Source: IGC (2020); Addis Fortune (2018, 2021); NUCA city profile; Egypt Ministry of Housing (Jan 2025); EPZA (2024); KenInvest; Rendeavour; Tatu City (2018, 2021); Le Matin (Oct 2025); IndustriALL (2014); Medias24 (2019); Gouvernement du Bénin (2023); India Infra Hub (2024). **Heirstone analysis.** Jobs and housing come from different dates; ratios are indicative. One home houses several people.

How the argument runs

Chapter 1 sets out the gap and the Heirstone Worker Housing Benchmark. Chapter 2 shows how the gap returns to the factory as turnover, using Ethiopia, the one African country where housing, rent, wages and turnover are documented together. Chapter 3 compares the bus with the bed. Chapter 4 shows why some dormitories fill and others stay empty. Chapter 5 prices a bed against the wage and reviews ten ways of delivering it. Chapter 6 turns to individual countries in North, West, East and Southern Africa. Chapter 7 sets out a test for new parks and recommendations by audience.

Turnover data exist only for Ethiopia; no other park in the benchmark publishes them. That is itself a finding: the measure that would show the cost of the gap is not collected.

The Heirstone bed test in brief

1. **Where will the workforce come from?** Hawassa: 91% from outside the city.
2. **Can a bus reach the labour pool?** Kenitra: 10 km, 480 minibuses.
3. **How many beds per 100 jobs, phase by phase?** Hawassa 7; Tamil Nadu 47.
4. **Does 20% of the entry wage cover the bed's capital?** Above US$68-220 a month, on assumed finance terms.
5. **Will the room beat the private market?** Hawassa tenants had 5.3 m² each.
6. **Who signs the lease and who runs it?** Employers and an operator.
7. **Is the land reserved in the masterplan?** China allows up to 15%.

CHAPTER 1



01

Jobs counted, beds not

Park operators publish sheds, investors and jobs. None of those in our benchmark publishes the number of beds available to the people who fill the jobs.

Africa's industrial parks are sold on serviced plots, ready-built sheds, power, customs and tax incentives, and their scorecards follow. Ethiopia's Industrial Parks Development Corporation (IPDC) reports more than 100,000 jobs across 13 parks and over 200 investors. Egypt's SCZone counts 212 operating plants at Sokhna and more than 133,000 direct jobs. Benin's Glo-Djigbé Industrial Zone (GDIZ) passed 25,000 jobs in 2026 and aims for 300,000 by 2030. Kenya's EPZ Authority reports 72,034 jobs across 101 zones.

None of these figures says where the workers sleep. Housing appears, when it appears at all, as a plan or a study: a 7,000-home estate on Kenya's investment portal, a study of worker housing announced for GDIZ in December 2023. Egypt's industrial new towns take a different route, with housing built city-wide: 10th of Ramadan has 194,133 homes alongside its 353,656 factory jobs.

That would matter less if workers lived nearby. Often they do not. The International Growth Centre (IGC) estimated that 91% of the 33,199 people working at Hawassa in April 2020 came from outside the city. In Tamil Nadu, 80% of the roughly 40,000 people at Foxconn's Sriperumbudur plant are from outside the area, according to India Infra Hub. Plants hiring at this scale recruit most of their staff from elsewhere, and those workers have to find somewhere to live.

WHERE THE ZONES ARE

From Tangier to Maseru, the zones that report their jobs rarely report where the workers live

EXHIBIT 2

Seventeen zones, four ways of housing the workforce

Zones in the Heirstone Worker Housing Benchmark, by the main way their workers are housed; circle size shows reported jobs (latest year available)



Source: Heirstone Worker Housing Benchmark 2026 (sources on pages 9 and 10). Positions of Hawassa and Bole Lemi from IPDC maps, and of 10th of Ramadan, Tatu City and Diamniadio from OpenStreetMap; other positions are approximate. Kenya EPZ jobs are national; the Athi River estate is shown at its approximate site. Mauritius and Lesotho are shown at national level. IPDC parks other than Hawassa and Bole Lemi, Naivasha and Renault Tanger Med are not plotted.

Only two of the zones on the map have dedicated worker beds in any number, and neither has them at scale. Hawassa reached about seven beds per 100 jobs. In Mauritius employer dormitories house migrant textile workers, who numbered about 15,000 of 65,000 textile workers in 2014.

Egypt has the two largest zones on the map by reported jobs, and its industrial areas, like Senegal's, sit inside new cities with general housing that is not tied to an employer. Morocco's Tangier and Kenitra zones bus their workers in. Elsewhere, from Bole Lemi to Lesotho, workers rent privately; for Kigali and GDIZ we found no data on where workers live.

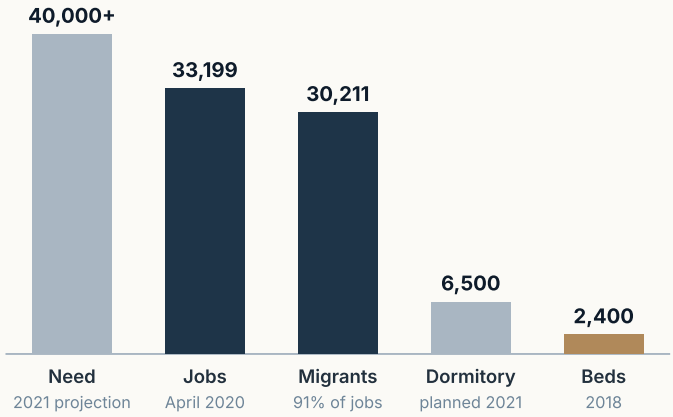
HAWASSA

At Hawassa, one dedicated bed for about every 13 workers who came from outside the city

EXHIBIT 3

Beds covered under a tenth of Hawassa's migrant workforce

Workers, accommodation need and dedicated beds at Hawassa Industrial Park, 2018 to 2021



Source: IGC (Haas, Kriticos and Lippolis, Nov 2020; Dercon, Caria and Lippolis, Sept 2019); Addis Fortune (Apr 2018; Sept 2021); FBC (Sept 2021). **Heirstone analysis.** 91% of 33,199. Beds are 600 IPDC-financed homes with up to four residents each.

Hawassa is the most fully documented park in our benchmark, and its numbers set the scale of the problem. The park opened in 2016 at a cost of US$245m, according to the IGC, or US$250m according to press reports at the opening. By April 2020 it employed 33,199 people, against a target of 60,000 working two shifts. The IGC estimated that 91% of them had come from outside the city.

Against that, IPDC financed 600 homes for up to four residents each, occupied by April 2018 according to its branch manager at the park: up to 2,400 beds, or one for every 13 workers from outside Hawassa. The government itself projected that more than 40,000 workers would need accommodation by 2021.

The response came in 2021: a 13-block dormitory village for 6,500 of the park's workers, budgeted at ETB 600m and backed by a KfW grant. If it was delivered, Hawassa would reach about 27 beds per 100 jobs. We could not verify its completion.

EXHIBIT 4

Worker housing announcements across Africa, and the latest public evidence of delivery

Selected announcements of housing for zone workers and national programmes in Africa, and the latest public evidence we found, 2013 to 2026

DATE	COUNTRY	ANNOUNCED	LATEST PUBLIC EVIDENCE
2013	Morocco	Chrafate new town near Tanger Med planned for 150,000 people, partly for car-plant workers	~420 homes delivered by October 2025; 418 more launched
May 2015	Egypt	World Bank approves US$500m for Egypt's Social Housing and Mortgage Finance Fund, followed by US$500m more in 2020	583,008 mortgages facilitated by June 2024; 28,140 social homes built or under way at 10th of Ramadan (Jan 2025)
2018	Ethiopia	IPDC finances 600 homes at Hawassa, up to four residents each, to cut turnover	Up to 2,400 beds against 33,199 jobs (2020)
Nov 2019	Kenya	EPZ Authority invites joint-venture bids for 5,000 homes at Athi River	Now marketed as a ~7,000-home, US$169m estate
Sept 2021	Ethiopia	Work starts on a 6,500-resident worker dormitory at Hawassa, completion targeted for mid-2023	Completion and occupancy not verified
Dec 2023	Benin	Government announces studies for health centres, schools and worker housing at GDIZ	25,000+ jobs by August 2026; no housing reported
Mar 2024	Kenya	Affordable Housing Levy collected at 1.5% of salary from workers and 1.5% from employers	KSh 73.2bn raised and 1,795 homes completed in the year to June 2025

Source: Medias24 (Oct 2013); Le Matin (Oct 2025); World Bank Projects (P150993, P168582); CAHF, Housing Finance in Africa Yearbook (2024); Egypt Ministry of Housing (Jan 2025); Addis Fortune (Apr 2018; Sept 2021); FBC (Sept 2021); Business Daily Africa (Nov 2019); KenInvest; Gouvernement du Bénin (Dec 2023); AllAfrica (Aug 2026); Kenya Revenue Authority (2024); Eastleigh Voice citing the Affordable Housing Board (Nov 2025).

THE BENCHMARK

The Heirstone Worker Housing Benchmark

Seventeen African zones and two international references. Jobs, housing and pay come from the latest public source for each item, so dates differ within a row. Beds or homes per 100 jobs is Heirstone analysis. n/p: not published or not found. n/a: not applicable. HFAE: Housing for All Egyptians. US$ at World Bank annual average rates (Benin and Nigeria 2025) or as given by the source (Ethiopia, Mauritius, Bangladesh 2017 rate).

EXHIBIT 5

Housing, access, pay and turnover at African industrial zones

ZONE	JOBS (DATE)	DEDICATED WORKER HOUSING	BEDS OR HOMES PER 100 JOBS	GETTING TO WORK	ENTRY PAY AND RENT	TURNOVER
ETHIOPIA						
Hawassa Industrial Park	33,199 (Apr 2020); target 60,000	Up to 2,400 beds in 600 IPDC-financed homes (2018); 6,500-resident dormitory begun 2021, completion not verified	7 (27 if dormitory delivered)	91% from outside the city; walks of up to 3 hours a day	Base 750 birr (US$26, 2019); shared room 31-37% of base; whole room up to 200%	5-10% a month; about 100% in year one
Bole Lemi Industrial Park	14,000 (Apr 2018); target 25,000	None reported	0	Inside Addis Ababa; private rentals near the park	Base 750 birr; rent near park ~900 birr (120%), per Addis Fortune citing IPDC	7,840 exits Jul-Dec 2017
IPDC parks (13)	100,000+ (c.2025); 57,000 (Sept 2022)	Hawassa projects only	n/p	n/p	No national private-sector minimum wage	~5,000 laid off after AGOA suspension (2022)
KENYA						
EPZ programme (101 zones)	72,034 (Mar 2024)	5,000-home tender (2019); ~7,000-home Athi River estate marketed (US$169m)	9.7 homes if built	n/p	n/p	n/p
Tatu City SEZ	30,000+ (developer); 25,000+ (press, Jul 2026)	7,000+ residents (developer); 1,640 affordable homes in two schemes, pipeline (announced 2018-21)	23 residents; 5.5 affordable homes	n/p	n/p	n/p
Naivasha SEZ	3,000+ committed by 2025 licensees	None reported	0	n/p	n/p	n/p
EGYPT						
10th of Ramadan City	353,656 (NUCA, undated)	City-wide: 194,133 homes built; 35,763 social and HFAE homes built or under way	55 homes; 10 social	49.3 km from Cairo; ~650,000 residents (NUCA estimate)	EGP 7,000 private minimum (2025)	n/p
Sokhna Industrial Zone (SCZone)	133,000+ direct (Jun 2026); 212 plants (Sept 2026)	20 km² residential area in 2018 plan; delivery not yet reported	n/p	n/p	EGP 7,000 private minimum (2025)	n/p
MOROCCO						
Tanger Automotive City	40,000+ direct (Sept 2024)	Chrafate new town: ~420 homes delivered by Oct 2025, shared with the wider zone	1.3 homes (per 33,000 zone jobs)	Employer buses; EUR 25 a worker a month to the Tanger Free Zone (2013)	Renault operators ~EUR 240 (2012); minimum MAD 3,111 (2023)	n/p
Renault Tanger Med	4,000+ (2013); 6,000 planned	No employer housing reported	0	35 km from Tangier; employer buses	~EUR 240 a month (2012)	n/p
Atlantic Free Zone, Kenitra	25,300 (Jul 2019); 30,000 (CDG)	None; 480 minibuses running in shifts	0 beds; 53 workers per minibus	10 km east of Kenitra	n/p	n/p

Source: Heirstone Worker Housing Benchmark 2026; sources listed on the following page.

THE BENCHMARK, CONTINUED

ZONE	JOBS (DATE)	DEDICATED WORKER HOUSING	BEDS OR HOMES PER 100 JOBS	GETTING TO WORK	ENTRY PAY AND RENT	TURNOVER
WEST AFRICA						
Glo-Djigbé Industrial Zone (GDIZ), Benin	25,000+ (Aug 2026); target 300,000 by 2030	Study of worker housing announced Dec 2023	0 verified	45 km north of Cotonou	Minimum CFA 52,000 (US$89)	n/p
Diamniadio and APROSI sites, Senegal	11,817 across APROSI sites	New city: 45,000 homes for 350,000 residents planned	n/a	30 km from Dakar by toll road	n/p	n/p
Lekki FTZ and Dangote refinery, Nigeria	8,000+ engineers expected (2017)	Construction camps for 22,000 and 10,000; ~1,000 staff in third-party estates (2017)	n/a	n/p	Minimum ₦70,000 (US$46)	n/p
EAST AND SOUTHERN AFRICA, ISLANDS						
Kigali SEZ, Rwanda	20,000+ (Apr 2026); 230+ industries	None reported	0	n/p	n/p	n/p
EPZ textiles, Mauritius	65,000 incl. 15,000 migrants (2014)	Employer dormitories for migrant workers	c.23 migrant beds	On or near site	MUR 6,000 (US$190, 2014)	n/p
Garment sector, Lesotho	~46,500 (2017)	None reported	0	Taxi or walking; transport about equal to rent	M1,600-1,800 (2019); two rooms M600-800	n/p
INTERNATIONAL REFERENCES						
Vallam Vadagal (SIPCOT), Tamil Nadu	~40,000 at Foxconn Sriperumbudur; 80% non-local	18,720 beds, six to a room, opened Aug 2024; a press report puts Foxconn's own blocks at 18,000 more	47 (92 with Foxconn's own, reported)	Industrial-park housing for the plant	Not disclosed	n/p
Ashulia dormitory, Bangladesh	Dhaka garment belt	744 beds, 62 to a room; no tenants after a year (2017)	n/a	n/p	Tk 1,000-1,200 a bed (US$12-15)	n/a

Source: Heirstone Worker Housing Benchmark 2026. IGC (2019, 2020); NYU Stern Center for Business and Human Rights (May 2019); Addis Fortune (Apr 2018); IPDC; The Reporter Ethiopia (Sept 2022); EPZA (2024); KenInvest; Business Daily Africa (2019); Rendeavour; Capital FM (Jul 2026); Tatu City (2018, 2021); African Business (2023); ADM (2019); Centre for Public Impact (2018); NUCA; Egypt Ministry of Housing (Jan 2025); Amwal Al Ghad (2018); Egypt Today (2026); Enterprise (Sept 2026); Challenge.ma (2024); Medias24 (2013, 2019); Le Matin (2025); Finances News Hebdo; L'argus (2012); CDG; Gouvernement du Bénin (2023); AllAfrica (2026); APROSI; ITB Nigeria; Sweetcrude Reports (2017); IndustriALL (2014, 2017); Public Eye (Lesotho, 2019); India Infra Hub (2024); The Realty Today (2025); The Daily Star (2017); minimum wages from Cairo Governorate (2025), Medias24 (2023), Agence Ecofin (2022), ThisDay (2024), The Reporter Ethiopia (2024) and Addis Fortune (2025); World Bank exchange rates.

READING THE BENCHMARK

Four patterns across seventeen zones

The benchmark is the starting point for the rest of the report. It sets each zone's jobs against whatever housing serves its workers, how they reach the gate, what they earn and pay, and how often they leave.

It is also a list of what is missing. In a well-run park, every cell in this table would be filled from the operator's own annual report.

Jobs are published; beds are not.

Every zone reports jobs, and most report investors and plants. None publishes a bed count alongside its jobs. The few housing figures come from press reports, research institutes and investment portals.

Housing follows jobs with a lag.

GDIZ announced a housing study in December 2023 with more than 10,000 jobs already on site; Kenya's Athi River estate has been offered since 2019.

Where housing exists it is mostly for households.

Egypt's and Senegal's new cities and Kenya's Tatu City are built around homes for sale to households; a rental product for workers in their first year would complement them.

Turnover is measured in only one country.

Ethiopia is the only African case where turnover can be set against rents and housing. No other park in the benchmark publishes turnover.

CHAPTER 2

02

Turnover is a housing bill

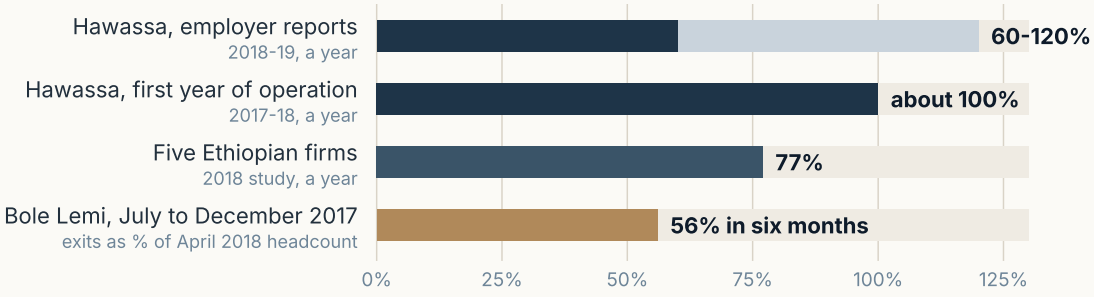
Ethiopia's first parks are usually described as a wage story. The base wage at Hawassa was 750 birr a month in 2019, about US$26 and 40% of national income per head, and NYU Stern's Center for Business and Human Rights set it beside turnover close to 100% a year.

The wage was only part of what workers faced. The same report found rooms costing up to twice the base wage, young women living four to a room without indoor plumbing, and walks of three hours a day where buses did not reach. Pay, rent and the journey to work all bore on whether workers stayed.

EXHIBIT 6

Ethiopian parks replaced most of their workforce every year

Staff leaving as a share of the workforce, Ethiopian industrial parks and firms, 2017 to 2019



Source: NYU Stern Center for Business and Human Rights, Made in Ethiopia (May 2019), including Blattman and Dercon (2018); Addis Fortune citing IPDC (Apr 2018).
Heirstone analysis. Bole Lemi: 7,840 exits against 14,000 employees in April 2018. Monthly rates of 5 to 10% equal 60 to 120% a year.

At 5 to 10% a month, a plant replaces its entire workforce every 10 to 20 months. Bole Lemi, inside Addis Ababa, lost more than 7,840 workers in the second half of 2017, according to IPDC figures reported by Addis Fortune, equal to 56% of the 14,000 it employed the following April, or about 9% a month. Hawassa lost close to 4,470 in the same six months.

IPDC itself made the link to housing. In 2018 it financed 600 homes at Hawassa, each for up to four residents paying 250 birr a month, to cut turnover, according to its Hawassa branch manager quoted by Addis Fortune. Workers interviewed by NYU Stern named the three-hour daily walk as a reason for quitting.

Turnover on this scale shows up in output. NYU Stern found factory efficiency at Hawassa ranging from 15% to 75%, and one large plant running at 25 to 30% in December 2018 that was not breaking even. Low efficiency has many causes, and the constant training of new operators is one of them.

> *Hawassa workers named a three-hour daily walk to and from the factory as a reason for leaving.*

10-20

months: the average stay of a worker when 5 to 10% leave every month

15-75%

range of factory efficiency at Hawassa reported in 2019

600

homes financed by IPDC at Hawassa in 2018 to cut turnover, at 250 birr a resident

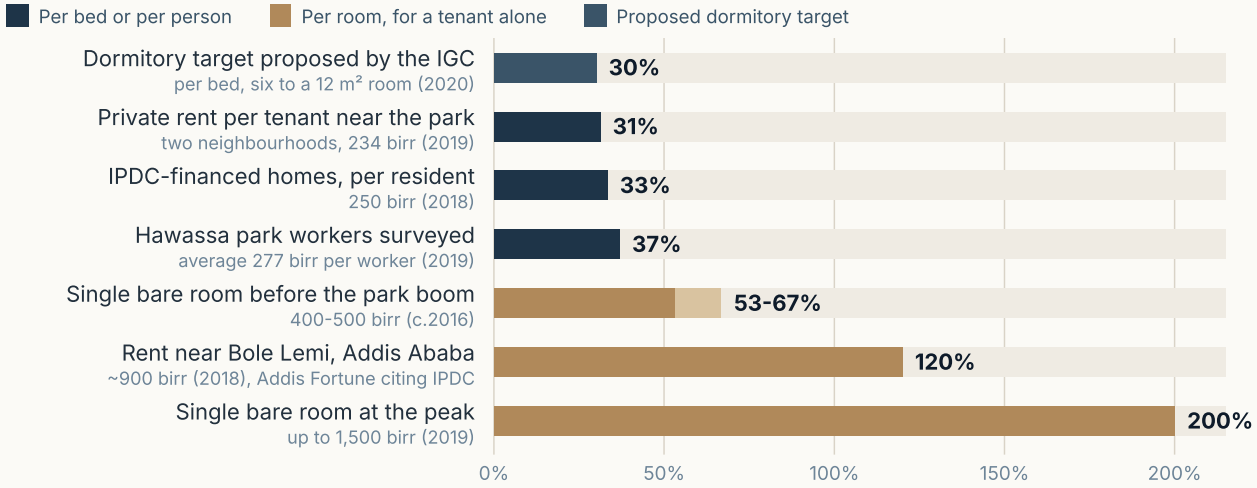
RENT AND WAGE

Per person, rents near Hawassa were already close to a dormitory target. Per room, they were out of reach.

EXHIBIT 7

A shared room cost a third of the base wage; a room of one's own cost up to twice the wage

Monthly rent as a share of the 750-birr entry-level base wage at Hawassa Industrial Park (Bole Lemi shown for comparison)



Source: NYU Stern CBHR, Made in Ethiopia (May 2019); IGC working paper, Abebe, Caria and Dercon (2019); IGC blog, Haas, Kriticos and Lippolis (Nov 2020); Addis Fortune (Apr 2018). **Heirstone analysis.** All rents divided by the 2019 base wage of 750 birr; rents span 2016 to 2019.

The informal market did house most of Hawassa's workers, and per person it did so at about a third of the base wage. In 2019 tenants in two neighbourhoods near the park paid an average of 234 birr a month, and park workers surveyed paid 277. The typical rented unit was one room shared by three people, with a median of 5.3 m² each. More than 85% of tenants rented from landlords who owned a single house.

The IGC found nominal rents up 26% from 2016 to 2019, against consumer prices up 33%, so real rents were flat or falling on average; a later IGC note found real increases of 20 to 43% in the segment most likely to house park workers. NYU Stern reported a bare room rising from 400-500 birr to as much as 1,500.

On an entry wage a worker can pay for a share of a room but not for a room alone. Workers therefore share with two or three others, or look further out, in neighbourhoods the buses do not reach. For the worker the cost shows up as crowding and distance more than as rent, and for the factory as departures. So a dormitory has to compete on location and space as well as on price.

5.3 m²
median floor space per tenant in rented rooms near the park (2019)

3
people sharing the typical rented room near Hawassa

85%
of tenants rent from landlords who own only one house

THE ARITHMETIC

Eight to 17 hires a year for every dedicated bed

At 60 to 120% a year, Hawassa's 33,199 jobs meant 20,000 to 40,000 hires a year. Against up to 2,400 dedicated beds, that is 8 to 17 hires for every bed. The question for a manufacturer is what a bed costs, and how much turnover it has to prevent to pay for itself.

The Hawassa dormitory begun in 2021 was budgeted at ETB 600m for 6,500 residents: about 92,300 birr, or US$2,111, per resident at the 2021 rate. On assumed finance of 6% over 25 years, its capital charge is US$13.8 a month per bed; at 10% over 20 years, US$20.7. Both exclude land and running costs. A rent of 20% of the 2019 base wage is US$5.2. The difference has to be met by the employer, a grant or cheaper land.

Exhibit 8 turns the gap into a test a plant manager can apply. Each cell shows how much an avoided departure must cost the factory, in recruitment, training and lost output, for the bed subsidy to pay for itself. If a bed cuts annual turnover by 40 points, from 100% to 60%, and each departure costs more than US$260 to 500, the subsidised bed is the cheaper option. Plant managers know their own cost of a departure; the grid lets them set it against the cost of a bed.

20,000-40,000

hires a year implied by Hawassa's turnover on 33,199 jobs

US$2,111

budgeted capital cost per resident of the Hawassa worker dormitory (ETB 600m, 6,500 residents, 2021 rate)

+17%

rise in take-home pay for a worker moving from a 277-birr shared room to a bed at 20% of base wage

EXHIBIT 8

At Hawassa a bed needs a subsidy; it pays if it prevents departures that cost a few hundred dollars each

Annual subsidy per bed and the break-even cost of one avoided departure, by rent cap, finance terms and cut in annual turnover (US$)

HAWASSA: US$2,111 BED (2021 BUDGET)	FINANCE	SUBSIDY A BED A YEAR (US$)	AS % OF ANNUAL BASE WAGE	TURNOVER CUT 20 POINTS	40 POINTS	60 POINTS
Rent at 20% of base wage	6%, 25 years	103	33%	US$514	US$257	US$171
Rent at 20% of base wage	10%, 20 years	186	59%	US$928	US$464	US$309
Rent at 15% of base wage	6%, 25 years	118	38%	US$592	US$296	US$197
Rent at 15% of base wage	10%, 20 years	201	64%	US$1,006	US$503	US$335
EGYPT FOR COMPARISON: EGP 7,000 MINIMUM WAGE (US$142 AT THE 2025 AVERAGE RATE)						
Rent at 20% of wage, US$2,100 bed	6% or 10%	0	0%	none needed	none needed	none needed
Rent at 20% of wage, US$4,500 bed	6%, 25 years	11	1%	US$54	US$27	US$18

Source: Heirstone analysis. Inputs: dormitory capex ETB 600m for 6,500 residents (Addis Fortune and FBC, Sept 2021) at ETB 43.73 per US$ (World Bank, 2021); base wage 750 birr = US$26 a month (NYU Stern, 2019); Egypt minimum wage EGP 7,000 (National Wages Council, 2025) at EGP 49.23 per US$ (World Bank, 2025). Finance terms are illustrative assumptions. Capital only: land, operations, maintenance and vacancy are excluded, so true subsidies are higher. Wage and cost years differ and the birr has since lost further value.

The same bed is also a pay rise. A worker paying 277 birr for a third of a room who moves to a bed at 20% of the base wage, 150 birr, keeps 127 birr more each month, 17% of her base wage, and gets back the hours she spent walking. Workers interviewed by NYU Stern cited both low pay and the long walk as reasons for leaving.

CHAPTER 3

03 The bus or the bed

Morocco built a car and components cluster of more than 150 suppliers and 40,000 direct jobs at Tangier without building worker housing next to it. It bused its workers in. That works where a city of renters sits within reach, and it has a price.



Kenitra's Atlantic Free Zone had 25,300 direct jobs and 25 operators in July 2019, above its 2009 target of 20,000, with under 45 hectares left. Its workers were moved by a fleet of 480 minibuses running in shifts, about 53 workers per vehicle. The Caisse de Dépôt et de Gestion (CDG), which developed it, places the zone 10 km east of Kenitra.

Tangier followed the same model at larger scale. Tanger Automotive City had more than 40,000 direct jobs in 2024, when the government approved its extension from 517 to 1,185 hectares. In 2013 employers paid about EUR 25 a worker a month to bus staff from Tangier to the free-zone factories. The Renault plant at Melloussa sits about 35 km from the city, according to Automotive Logistics. The Chrafate new town, launched near the zone in 2013, was expected to cut that transport cost by at least half.

The bus model fits Kenitra: a zone 10 km from a city with an existing rental market. It fits less well where the zone is far from any city or where the workers come from far away.

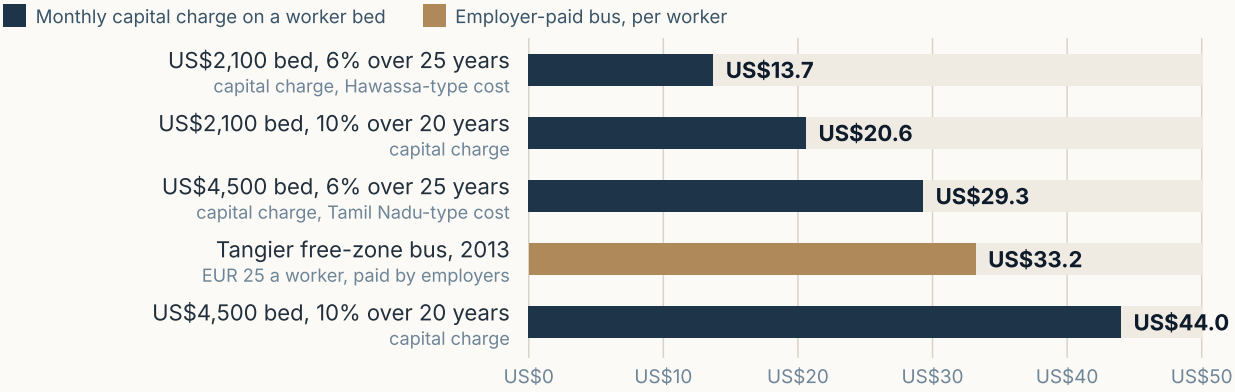
THE PRICE OF THE BUS

Tangier's monthly bus cost per worker would carry the capital charge on a dormitory bed

EXHIBIT 9

A bus seat and a bed's capital charge cost about the same per worker each month

Employer bus cost per worker (Tangier free zone, 2013) and monthly capital charge on a worker bed, US$ a month



Source: Medias24 (Oct 2013); World Bank exchange rates (EUR 0.753 per US$, 2013). **Heirstone analysis.** Capital charge = capex x capital recovery factor / 12; finance terms are assumptions. The bus figure excludes the rent a bused worker still pays in town; the bed figures exclude land and operations.

At the 2013 average rate, Tangier's EUR 25 was about US$33 a month. The same amount would service the capital on a bed costing US$3,400 at 10% over 20 years, or US$5,100 at 6% over 25 years, before land and running costs, a range that spans the plain dormitories built in Ethiopia, Bangladesh and India. It cost employers the equivalent of about 10% of a Renault operator's 2012 wage of about EUR 240, as reported by L'argus, and the worker still paid rent in town.

Where the bus wins is flexibility. Routes can be added or cancelled as plants open, close or change shifts; a dormitory is a 20-year asset. The bus is the right answer when the labour pool already lives within reach, and the bed when it does not. Kenitra, 10 km from a city with a rental market, is a natural bus zone. A zone 45 km from the nearest city, or one hiring mostly from rural areas, is not.

Egypt's industrial new towns sit between the two. 10th of Ramadan, 49.3 km from Cairo, combines 353,656 factory jobs with 194,133 homes built city-wide and, on NUCA's estimate, about 650,000 residents (Exhibit 18). Its social housing is sold to households.

Lesotho shows the cost of having neither. Garment workers in Maseru earned M1,600 to 1,800 a month in 2019, according to Public Eye, a local newspaper, while a two-room rental on the outskirts cost M600 to 800 and transport almost as much again. On those figures, rent and fares together could take two thirds or more of pay.

> *For what Tangier's employers paid to bus one worker in 2013, a lender could finance a dormitory bed costing US$3,400 to 5,100, at 6 to 10% interest.*

53

workers per minibus at the Atlantic Free Zone: 25,300 workers, 480 minibuses (2019)

US$33

a month: Tangier's 2013 bus cost per worker at the 2013 average exchange rate

US$3,400-5,100

bed whose capital a US$33 monthly payment would service, at assumed rates of 10% and 6%

67-100%

of a Lesotho garment wage taken by a two-room rental and transport together, on Public Eye's 2019 figures

CHAPTER 4

04

Design decides who moves in



Two dormitories built for women factory workers cost almost exactly the same per bed. One stood empty after a year. The other opened with 18,720 beds. The difference lay in how the rooms were laid out and who ran the building.

Ashulia, in Dhaka's garment belt, put 62 beds in each room. Vallam Vadagal, near Chennai, put six in each and was handed to the anchor employer to run.

EXHIBIT 10

Same cost per bed, opposite results

Two state-built dormitories for women factory workers compared

	ASHULIA, DHAKA (BANGLADESH)	VALLAM VADAGAL, TAMIL NADU (INDIA)
Built by	Public Works Department	State of Tamil Nadu, through its industrial agency SIPCOT
Ready or opened	About late 2016	17 August 2024
Beds	744 women garment workers	18,720 women workers of Foxconn
Cost	Tk 26.98 crore	Rs 706.5 crore (Rs 706.05 crore in another report)
Cost per bed	US$4,508 (2017 rate)	US$4,511 (2024 rate)
Beds per room	62	6
Sanitation	12 toilets and 12 bathrooms for 744 beds	Not disclosed
Layout	Barrack-like rooms	13 blocks, stilt plus ten floors, 24 rooms a floor, on 20 acres
Who runs it	Directorate of Women Affairs lets beds one by one	Foxconn runs housekeeping and security; SIPCOT the infrastructure
Rent	Tk 1,000 a bed; Tk 1,200 for single mothers (US$12-15)	Not disclosed
Outcome	No tenants after about a year (Nov 2017)	Opened with 18,720 beds; occupancy not disclosed

Source: The Daily Star (Nov 2017); India Infra Hub (Aug 2024); Deccan Herald (Aug 2024). **Heirstone analysis.** Cost per bed at World Bank average rates: Tk 80.44 per US$ (2017), Rs 83.67 per US$ (2024).

Bangladesh's Public Works Department built Ashulia for Tk 26.98 crore and handed it to the Directorate of Women Affairs to run. In November 2017, when it had been ready for almost a year, The Daily Star reported that it had no tenants. Rent was Tk 1,000 a bed, about US$12, at a time when Apparel Resources reported garment workers typically spending about half their income on housing. The Daily Star pointed instead to the barrack-like rooms of 62 beds and the lack of a gas connection; 12 toilets served 744 beds.

Tamil Nadu took the opposite route. Its industrial agency built 13 blocks of six-bed rooms on 20 acres and handed housekeeping and security to Foxconn, whose Sriperumbudur plant employs about 40,000 people, 80% of them from outside the area. The anchor employer needed the beds and runs them; the state carried the capital. India's 2024-25 budget then committed to rental dormitories for industrial workers through public-private partnerships with viability gap funding and an anchor-industry commitment, the same two features.

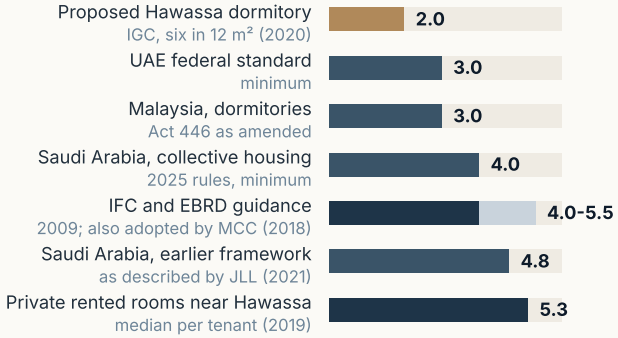
STANDARDS

Standards differ by almost two to one on space, and the cheapest designs fall below the private market

EXHIBIT 11

A worker in a private room near Hawassa had more space than the proposed low-cost dormitory offered

Floor space per worker, m²: standards and practice

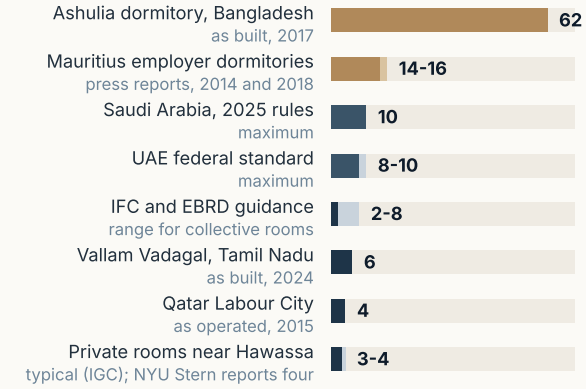


Source: IFC and EBRD (2009); MCC (2018); UAE government portal; Malaysia Act 446 regulations (secondary compilation); Gulf News and Gulf Business (2025); JLL (May 2021); IGC (2019, 2020).

EXHIBIT 12

Rooms of more than eight people fall outside every lender standard

Persons per room: standards and practice



Source: The Daily Star (Nov 2017); IndustriALL (2014, citing press); Dagens Arbete (2018); Gulf News (2025); UAE government portal; IFC and EBRD (2009); India Infra Hub (2024); Khaleej Times (2015); IGC (2019); NYU Stern (2019).

The low-cost dormitory proposed for Hawassa in 2020 put six workers in a 12 m² room for about 30% of the base wage. That is 2 m² each, about 38% of the 5.3 m² a typical tenant already had in a private room for about the same rent. A worker comparing the two would have little reason to move. A dormitory has to beat the room it replaces on the things the worker values, and space and privacy come first.

Lender standards settle the design for many exporters. IFC and EBRD guidance cites benchmarks of 4 to 5.5 m² per worker, two to eight people per room, no triple bunks and sanitation from one unit per six to one per 15 people. Building to that band instead of the 3 m² minimum in UAE and Malaysian rules needs 33 to 83% more sleeping space per bed. Any park with a DFI-financed tenant, or a tenant selling to brands that audit their suppliers, should design to the stricter band from the start.

Saudi Arabia has moved the other way on density while widening enforcement. Its 2025 rules allow up to ten people per room at 4 m² each, against four per room at 4.8 m² in an earlier framework described by JLL, and in March 2026 the authorities pressed firms with 20 or more workers to license their housing, according to People Matters ME. In Mauritius, press reports in 2014 and 2018 described employer dormitories with 14 to 16 people to a room.

38%

of the private market's floor space per worker offered by the proposed Hawassa dormitory design

33-83%

more sleeping area per bed under IFC and EBRD guidance than under a 3 m² minimum

1

person per room: South Africa's 2014 target for converted mine hostels

CHAPTER 5

05

What a bed costs, and who pays

Plain dormitories in Ethiopia, Bangladesh and India have been budgeted or built at US$2,100 to 4,500 a bed. Projects that add malls, clinics and stadiums cost three to six times as much.

3-6x

spread in cost per bed between plain dormitories and amenity-rich labour cities

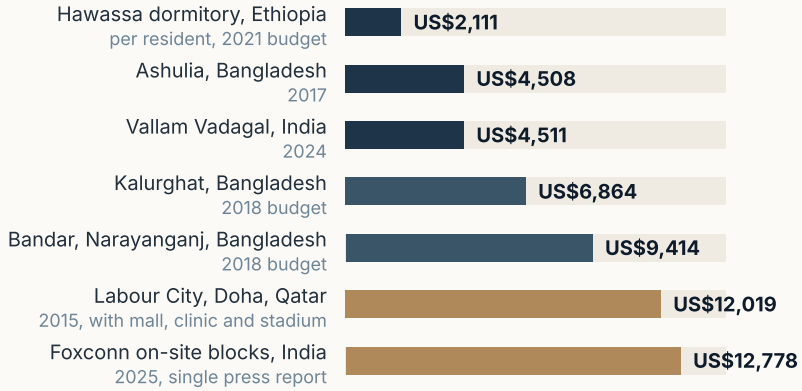
4.3 m^2

of land per bed at Vallam Vadagal: 18,720 beds on 20 acres

EXHIBIT 13

The building costs little; the amenities around it multiply the bill

Capital cost per bed or resident of named worker housing projects, US$ at World Bank average rates for the cost year



Source: Addis Fortune and FBC (Sept 2021); The Daily Star (2017, 2018); India Infra Hub (2024); Khaleej Times (2015); The Realty Today (Jan 2025). **Heirstone analysis.** Reported cost divided by reported beds or residents; Hawassa is per resident. Qatar and Foxconn on-site figures include non-residential facilities and may not be like for like.

The capital cost of a plain bed is modest. Ethiopia's Hawassa dormitory was budgeted at about US$2,100 per resident; Bangladesh's Ashulia and Tamil Nadu's Vallam Vadagal both to about US$4,500 a bed. Two later Bangladesh dormitories were budgeted at US$6,900 and 9,400 a bed. Qatar's Labour City, which opened in 2015 with 68,640 beds, a mall, a clinic and a 24,000-seat cricket stadium, cost about US$12,000 a bed, and Foxconn's own blocks at Sriperumbudur about US$12,800, according to a single trade-press report (The Realty Today).

The land needed is small. Vallam Vadagal fits 18,720 beds on 20 acres, about 2,300 beds a hectare. At that density every one of Hawassa's 33,199 workers could be housed on 14 hectares, under 5% of the park's 300 ha. At half the density, in lower blocks, it would take about 10%. On our estimate, housing all of Hawassa's workers from outside the city at the dormitory's budgeted cost would have come to about US$64m, about a quarter of what the park itself cost.

Standards move the cost as well. Building to the IFC and EBRD benchmark of 4 to 5.5 m^2 per worker, rather than a 3 m^2 minimum, means 33 to 83% more sleeping area per bed, before sanitation and dining space. Where employers already house workers, the bed is in the cost base: in 2008 a UAE contractor told MEED that accommodation took up to 25% of its labour costs, with another 25% going on transport, food and flights.

> *At Tamil Nadu's density, every worker at Hawassa could sleep on under 5% of the park's land.*

The harder question is who pays the gap between what the bed costs and what a worker can pay. That depends on the wage.

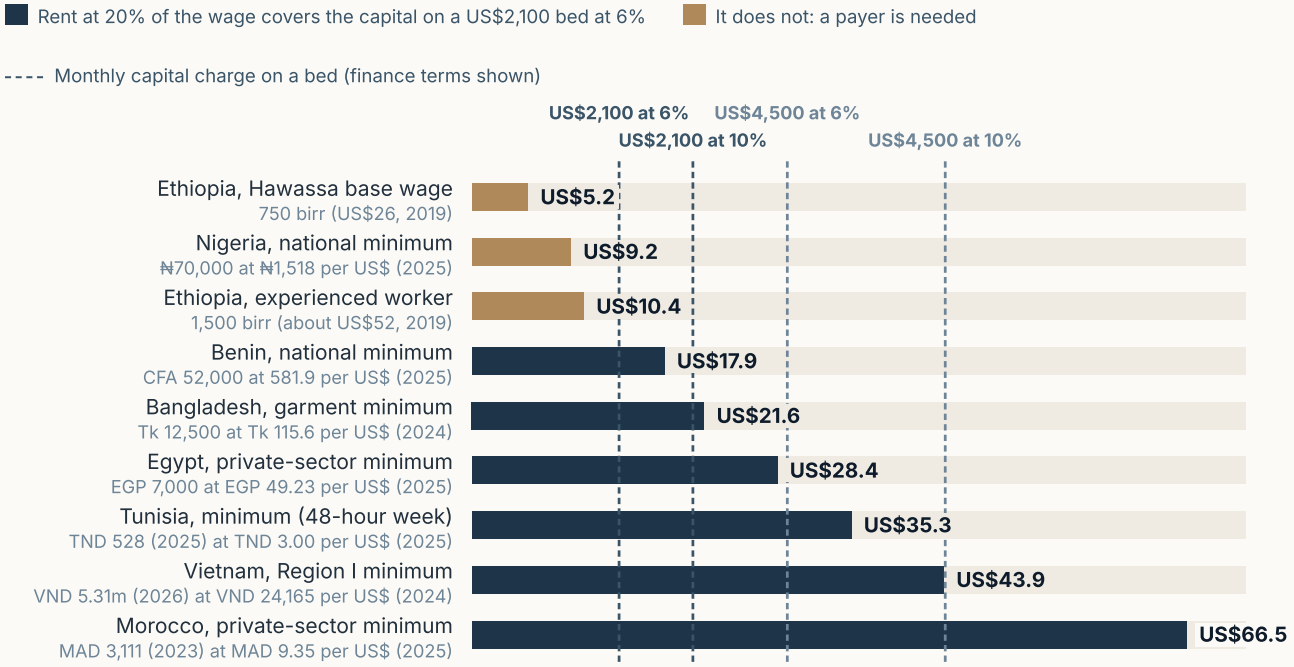
THE HEIRSTONE RENT-CAP MODEL

From an entry wage of US$68 to 220 a month, a bed let at a fifth of pay covers its capital charge

EXHIBIT 14

North Africa clears the bar; Ethiopia and Nigeria need someone to fill the gap

Rent at 20% of the minimum or base wage, US$ a month, against the monthly capital charge on a plain worker bed (assumed finance terms; land and operating costs excluded)



Source: Wages: NYU Stern (2019) for Ethiopia; ThisDay (2024) for Nigeria; Agence Ecofin (2022) for Benin; WageIndicator (Dec 2023) for Bangladesh; Cairo Governorate (2025) for Egypt; Managers (2024) for Tunisia; EY Vietnam (2025) for Vietnam; Medias24 (2023) for Morocco. Exchange rates: World Bank annual averages. **Heirstone analysis.** Capital charge = capex x capital recovery factor (0.0782 at 6% over 25 years; 0.1175 at 10% over 20 years) / 12; the finance terms are assumptions. Capital only: land, operations, maintenance and vacancy are excluded.

The model asks one question: if a worker pays 20% of the entry wage for a bed, does that cover the capital cost of building it? On assumed finance of 6% over 25 years, a plain bed at US$2,100 needs a wage of about US$68 a month; at 10% over 20 years, US$103. A bed at US$4,500 needs US$147 or US$220.

Egypt clears both US$2,100 bars and sits at the first US$4,500 bar; Tunisia clears three of the four and Morocco all four. Egypt's EGP 7,000 minimum gives a 20% rent of EGP 1,400, about US$28, which covers a US$2,100 bed at either rate and almost covers a US$4,500 bed at 6%. There the open questions are finance and operation.

Benin sits on the line: CFA 52,000 covers a US$2,100 bed at 6% but not at 10%. At Ethiopia's 2019 base wage and Nigeria's minimum, a fifth of pay covers between a quarter and two thirds of the capital charge on a US$2,100 bed. There a bed is a wage supplement, and its cost has to be met by employers, grants or land.

These are capital charges on assumed terms. Operating costs, maintenance and empty beds come on top, and we found no published operating cost per bed, so the thresholds are floors. Morocco's minimum is shown at its 2023 level; it rose by 5% in 2025 and again in 2026, so its bar is conservative.

US$68

monthly wage at which 20% of pay covers the capital charge on a US$2,100 bed at an assumed 6% over 25 years

64%

of a Hawassa worker's annual base wage: the yearly subsidy per bed needed to hold rent to 15% of pay at 10% finance

DELIVERY MODELS

Ten ways to deliver a worker bed, judged on what they have produced

Who pays for the building, who runs it and who pays the rent decide whether beds appear at all. The typology below sorts the cases in this report by those three questions. The final column is our judgement of how well each model fits a multi-tenant African industrial park, based on the outcomes shown.

EXHIBIT 15

The models that deliver beds put an employer on the lease and a professional operator in charge

Heirstone typology of worker housing delivery models, with named cases and outcomes

MODEL	WHO PAYS FOR THE BUILDING	WHO RUNS IT	WHO PAYS THE RENT	EXAMPLE	WHAT HAPPENED	FIT FOR AFRICAN PARKS
A State-built, employer-run	State or park agency	Anchor employer	Employer and workers	Vallam Vadagal, Tamil Nadu (2024)	18,720 beds opened for one anchor employer	Strong where an anchor employer exists
B Employer-built on site	Employer	Employer	Employer and workers	Foxconn Sriperumbudur; Mauritius textile firms	Foxconn blocks for 18,000 reported; Mauritius rooms of 14 to 16 in press reports	Strong for large single tenants
C Developer-built, employer-leased	Developer equity, grant and bank loan	Private developer	Employers, redirecting housing and transport allowances; workers in part	Hawassa dormitory, Ethiopia (2021)	6,500 residents planned; completion not verified	Strong on paper for multi-tenant parks
D Land incentive	Employers or park developers	Employers or operators	Workers, below market	Hawassa free land (2019); China 7% to 15% rule (2021)	At least three Hawassa suppliers interested; national policy in China	Strong, and costs little public money
E Mandate on employers	Employers or licensed providers	Licensed operators	Employers	UAE: 50+ workers under AED 1,500 (2022); Saudi licensing push (2026)	Employer-paid demand for licensed operators	Conditional on enforcement capacity
F State-built, let bed by bed	State	Government department	Workers	Ashulia, Kalurghat, Bandar, Bangladesh (2017-18)	Ashulia had no tenants after a year	Weak
G Ownership social housing and new towns	State and developers, with subsidy	Housing agency	Buyers	Egypt HFAE; Morocco Chrafate; Vietnam one-million homes plan	28,140 HFAE homes built or under way at 10th of Ramadan; 420 homes at Chrafate in 12 years	Serves households; first-job workers need a rental complement
H Payroll levy fund	All formal employers and employees	State housing board	Buyers or tenants	Kenya Affordable Housing Levy (2024)	KSh 73.2bn raised; 1,795 homes completed, year to June 2025	Strong if part is channelled to rental beds
I Mining employer housing	Mining companies, with state support	Mining companies	Employees, as tenants or buyers	South Africa hostel conversion (2010-14)	45% of right holders missed the target (government); ~90% converted (industry)	Lesson in measurement
J State labour city	State	Private operator	Employers	Labour City, Doha (2015)	68,640 beds at about US$12,000 a bed	Weak: costly amenities

Source: India Infra Hub (2024); The Realty Today (2025); IndustriALL (2014); Addis Fortune (Sept 2021); NYU Stern (2019); State Council of China, Guo Ban Fa [2021] No. 22; UAE government portal; People Matters ME (Mar 2026); The Daily Star (2017, 2018); Egypt Ministry of Housing (2025); Le Matin (Oct 2025); Viet Nam Government Portal (2025); Kenya Revenue Authority (2024); Eastleigh Voice (Nov 2025); gov.za (May 2015); Polity (Nov 2014); Khaleej Times (2015). **Heirstone analysis.** The typology and the final column are Heirstone judgements.

The models that produced beds share two features. An employer signs the lease, so occupancy does not depend on individual workers deciding one by one, and a professional operator, whether the anchor employer or a developer, runs the building to a standard. In each case the beds sit inside or next to the park.

Ownership models have neither feature. Morocco's Chrafate flats start at MAD 250,000, or MAD 150,000 after the state's direct housing aid: 48 months of the 2023 private-sector minimum of MAD 3,111. A young operator in her first job is unlikely to buy a flat, and the factory that needs her has no stake in the transaction.

WHERE THE MONEY WENT

Housing money has flowed to ownership and to zone platforms, and rarely to the bed at the factory gate

EXHIBIT 16

In the programmes shown, completed homes have trailed the money raised or committed

Selected housing programmes: funds raised or committed, and delivery

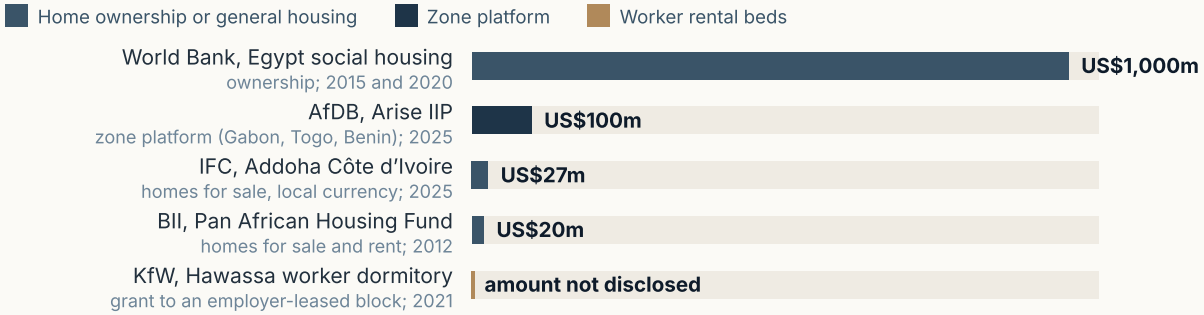
PROGRAMME	MONEY RAISED OR COMMITTED	WHAT WAS DELIVERED	DELIVERED RELATIVE TO PLAN OR FUNDS
Kenya Affordable Housing Levy	KSh 73.2bn in the year to June 2025 (US$566m at the 2025 rate)	1,795 homes completed in the year; over KSh 30bn held in Treasury bills	Over 41% of a year's collections held in Treasury bills
Vietnam social housing credit package	VND 120 trillion of low-rate State Bank credit	VND 416bn disbursed by February 2024, to five projects	0.35% of the package disbursed
Vietnam social housing, 2011-2020	National programme	About 100,000 apartments in 248 projects	41.4% of plan
Chrafate new town, Morocco	1,300 ha planned for 150,000 people (2013)	Just over 420 homes delivered by October 2025; 418 more launched	1.3 homes per 100 jobs in the zone

Source: Kenya Revenue Authority (2024); Eastleigh Voice citing the Affordable Housing Board (Nov 2025); Business Daily Africa (Jun 2026); VOV (Feb 2024); Viet Nam Government Portal (Mar 2025); Open Development Mekong (2023); Medias24 (Oct 2013); Le Matin (Oct 2025). **Heirstone analysis.** Shares computed from the figures shown; KSh converted at the World Bank 2025 average of 129.3 per US$.

EXHIBIT 17

We found one development finance deal in Africa ring-fenced for factory worker beds

Selected development finance commitments to housing and industrial zones in Africa, US$ million



Source: World Bank Projects (P150993, P168582); Africa Private Equity News (May 2025); IFC (Mar 2025); British International Investment; Addis Fortune (Sept 2021). The AfDB announcement did not earmark worker housing.

Kenya's levy shows how much money is already being collected. One year's collections, US$566m at the 2025 rate, equal the capital cost of 126,000 to 270,000 beds at US$2,100 to 4,500 each, more than one for every one of the 72,034 people in Kenya's EPZs. In June 2026 the Kenya Revenue Authority said employers had not remitted more than KSh 100bn of deductions. What is missing is a rental product next to the jobs for the levy to fund.

Development finance follows the products on offer: mortgages and homes for sale, and zone platforms that build sheds and power. IFC lent Addoha's Côte d'Ivoire business the local-currency equivalent of US$27m for 5,600 homes in Grand Abidjan, about 1,000 of them for low-income buyers. The only ring-fenced worker dormitory finance we found in Africa is KfW's grant to the Hawassa block, alongside 15% developer equity and a Zemen Bank loan.

126,000-270,000

beds whose capital cost equals one year of Kenya's housing levy collections

0.35%

of Vietnam's VND 120 trillion social housing credit package disbursed by February 2024

CHAPTER 6

06

North Africa and Sub-Saharan Africa

The same problem looks different in each market. North Africa pays wages that can carry the capital cost of a rental bed, and has so far housed workers through new towns, ownership flats and buses. Ethiopia has the clearest evidence of the cost of missing beds and the lowest wages to pay for them. West Africa's new zones are early enough to plan beds before the workforce arrives.

AT A GLANCE

Morocco	A fifth of the minimum wage covers the capital charge on any of the reference beds. Rental residences beside Tangier and Kenitra would fill a gap left by ownership flats.
Egypt	A fifth of EGP 7,000 covers the capital charge on a plain bed, and the industrial new towns have housing land. A licensed rental dormitory is the next step.
Tunisia	Wages a little above Egypt's; no park-level housing data published.
Benin and Senegal	GDIZ can still plan its beds; Benin's minimum covers a plain bed's capital charge at 6%.
Nigeria	At ₦70,000 a bed needs a payer, and the naira's fall argues for local-currency finance.
Ethiopia	The best evidence and the deepest affordability gap: grants and employer leases will be needed.
Kenya	The levy could fund rental beds in SEZs; the constraint is delivery.

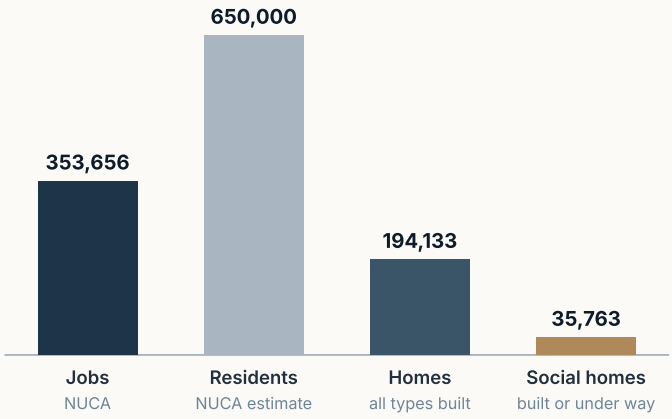
EGYPT

Egypt: industrial cities at scale, and a wage that can carry a bed's capital cost

EXHIBIT 18

10th of Ramadan City: 353,656 factory jobs and 194,133 homes built

10th of Ramadan City: factory jobs, residents and homes



Source: New Urban Communities Authority (NUCA) city profile (undated, retrieved 2026); Egypt Ministry of Housing (Jan 2025). **Heirstone analysis.** Social homes are 7,623 social housing units plus 28,140 Housing for All Egyptians units completed or under construction.

Egypt's industrial cities employ people on a large scale. NUCA counts 2,056 producing factories and 353,656 jobs in 10th of Ramadan City alone, 49.3 km from Cairo. SCZone's Sokhna zone reported more than 133,000 direct jobs and US$33.1bn of investment across 547 projects by mid-2026, with 212 plants operating in September 2026 and 176 under construction.

On NUCA's estimate of 650,000 inhabitants, 10th of Ramadan has about 1.8 residents per factory job, and 194,133 homes have been built city-wide, about 55 per 100 jobs. Its social homes are sold to households.

Sokhna's 2018 plan with DP World, for a 95 km^2 development, included a 20 km^2 residential area for 500,000 to 650,000 people, depending on the source. Progress on it has not yet been published.

Egypt's housing answer has been the new city and a social housing programme at national scale. The Social Housing and Mortgage Finance Fund had facilitated 583,008 mortgages by June 2024, and the World Bank has committed US$1bn to it since 2015. In 10th of Ramadan, 28,140 Housing for All Egyptians units were completed or under construction by January 2025, including a 3,000-unit cluster of 90 m^2 flats that was 98% complete in October 2024. These family homes for purchase serve households; a rental bed for a young machine operator who has moved to the city for her first job would complement them.

At the EGP 7,000 private-sector minimum set by the National Wages Council from 2025, US$142 a month at the 2025 average rate, a rent of 20% is EGP 1,400. That covers the capital charge on a US$2,100 bed at an assumed 6% or 10%, and 97% of it on a US$4,500 bed at 6%, before land and running costs. A plain bed needs little or no capital subsidy. The next step is a licensed rental product on industrial land, an operator, and employers willing to lease blocks.

A dormitory let in pounds is best financed in pounds, so that rent and debt service move together. Robbiki Leather City, whose 176-acre first phase has 216 tanning and storage units allocated to 192 investors (2024), is another site where published data on worker housing would help investors size demand.

133,000+

direct jobs at Sokhna by mid-2026, with 212 plants operating in September 2026 and 176 under construction

EGP 1,400

a month: 20% of the private minimum wage, enough to carry the capital charge on a plain bed at assumed finance terms

583,008

mortgages facilitated by Egypt's Social Housing and Mortgage Finance Fund by June 2024

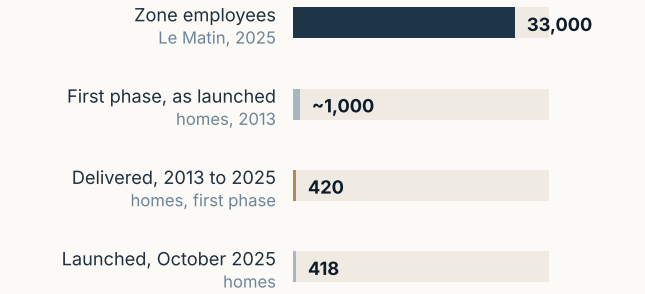
MOROCCO, TUNISIA AND ALGERIA

Morocco has relied on buses and starter flats. Its wages would support rental beds.

EXHIBIT 19

Chrafate delivered about one home for every 80 zone workers in twelve years

Chrafate new town near Tanger Med: zone employment and homes

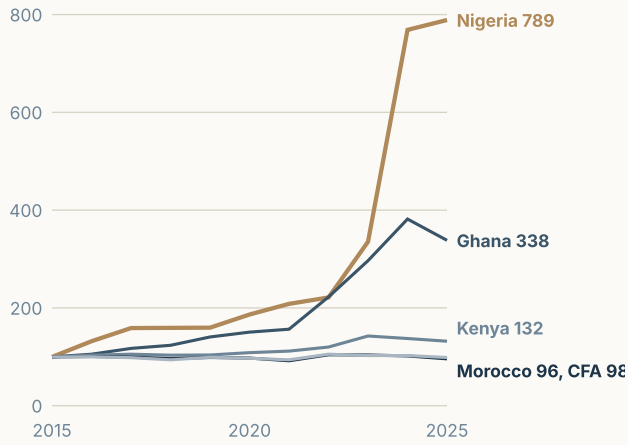


Source: Medias24 (Oct 2013); Le Matin (Oct 2025). The town was planned on 1,300 ha for 150,000 inhabitants.

EXHIBIT 20

Since 2015 the dirham and CFA franc have held their dollar value; floating currencies have adjusted

Local currency per US$, index 2015 = 100, annual averages



Source: World Bank, official exchange rate (PA.NUS.FCRF), retrieved 2026. **Heirstone analysis.**

Morocco's Tangier and Kenitra zones were built without worker beds. Kenitra's Atlantic Free Zone relied on 480 minibuses. Tangier's employers paid for buses, and the state's land developer Al Omrane launched Chrafate in 2013 to bring homes closer to the zone. Twelve years later it had delivered just over 420 homes against 33,000 employees in the industrial zone, and launched 418 more.

The wage supports a different product. The private-sector minimum of MAD 3,111 set in 2023, about US$333 at the 2025 average rate and raised twice since, puts a 20% rent at MAD 622, enough to cover the capital charge on any of the reference beds in this report at the finance terms assumed here. With Tanger Automotive City approved to grow from 517 to 1,185 ha, there is room for rental residences that employers lease for new hires. The dirham has held its value against the dollar since 2015, which makes long-term finance simpler.

Tunisia's minimum wage for a 48-hour week, TND 528 a month from 2025 before allowances, was about US$176 at the 2025 rate, a little above Egypt's. A fifth of that wage would cover the capital charge on a plain bed there too, and on a US$4,500 bed at an assumed 6%. We found no park-level data on worker housing in Tunisia or Algeria, so we draw no conclusions for them beyond this arithmetic.

A practical first step would be a block of rental beds for first-year operators beside Tanger Automotive City, on public land, leased by a group of suppliers and run by a private operator. Its published occupancy and turnover would show whether the bus is really the cheaper choice.

MAD 622

a month: 20% of Morocco's private-sector minimum wage

48

months of minimum wage to buy the cheapest Chrafate flat after state aid (MAD 150,000)

EAST AND SOUTHERN AFRICA

Ethiopia needs a payer for its beds. Kenya already collects one, in its housing levy.

Ethiopia

IPDC reported 57,000 jobs across its 13 parks in September 2022, against a target of 75,000, and more than 100,000 by about 2025. About 5,000 workers were laid off after the suspension of Ethiopia's AGOA trade preferences, and officials blamed turnover on swings in orders and low salaries. Ethiopia has no national private-sector minimum wage: the 2019 labour law provides for a wage board to set one, but none had been set by 2025, according to The Reporter Ethiopia and Addis Fortune.

The policy tools are already on the table. In 2019 the government offered manufacturers free land inside Hawassa if they built dormitories, and at least three suppliers, Arvind, Epic Group and Isabella, expressed interest. The Cornerstone dormitory launched in 2021 was to combine up to 15% developer equity, a KfW grant and a Zemen Bank loan, and to lease blocks to employers that already pay housing and transport allowances. That structure turns an existing cost into a lease. Its occupancy, once published, is the most useful single data point for worker housing in Africa.

The birr has lost much of its value (ETB 23.9 per US$ in 2017, 43.7 in 2021, on World Bank averages), so dollar-denominated costs and birr rents should not be compared across years without adjustment.

Kenya

Kenya's 101 EPZs employed 72,034 people in March 2024, up 9.1% in a year. Tatu City reports more than 30,000 jobs and just over 7,000 residents, and the EPZ Authority markets a 7,000-home Athi River estate at US$169m, about US$24,000 a home, on the grounds that investors need housing close to factories. That is the price of a family home, many times the cost of a worker bed. Kenya's housing levy, 1.5% of gross salary from employees matched by employers, is the country's advantage: if a share could fund employer-leased rental beds in SEZs, the money already exists.

Mauritius, Lesotho, South Africa and Rwanda

In Mauritius employer dormitories are common for migrant textile workers: about 15,000 of 65,000 in 2014, earning about MUR 6,000 a month. Press reports in 2014 and 2018 described rooms of 14 to 16, above the eight-person ceiling in IFC and EBRD guidance. Lesotho's 46,500 garment workers rent privately and, according to Public Eye, pay almost as much again for transport. South Africa's 2010 Mining Charter required mine hostels to become single rooms or family units by 2014; the government found 45% of right holders had not met it, while industry reported about 90% conversion. The gap between the two figures shows why any bed target needs an agreed definition. Kigali's SEZ hosts more than 230 industries and 20,000 jobs; we found no worker housing data for it.

57,000

jobs in Ethiopia's 13 IPDC parks in September 2022, against a 75,000 target

US$24,000

per home for the 7,000-home EPZA estate marketed at Athi River

KSh 100bn+

of housing levy deductions not remitted by employers, according to the Kenya Revenue Authority (June 2026)

45%

of South African mining right holders missed the 2014 hostel conversion target, according to the government

WEST AFRICA

West Africa's newest zones can still plan their beds before the workforce arrives

EXHIBIT 21

At its 2030 target GDIZ would need a worker town on a few per cent of its land

Illustrative scenario: worker beds, land and capital for Benin's Glo-Djigbé Industrial Zone at 300,000 jobs

SCENARIO AT 300,000 JOBS	BEDS	LAND AT VALLAM VADAGAL DENSITY (HA)	LAND AT HALF THAT DENSITY (HA)	SHARE OF GDIZ'S 1,640 HA	CAPITAL AT US$2,100-4,500 A BED
25 beds per 100 jobs	75,000	32	65	2-4%	US$158-338m
47 beds per 100 jobs (Tamil Nadu reference)	141,000	61	122	4-7%	US$296-635m
FOR COMPARISON: TODAY'S 25,000 JOBS					
47 beds per 100 jobs	11,750	5	10	under 1%	US$25-53m

Source: Heirstone analysis. Inputs: GDIZ target of 300,000 direct jobs by 2030 and 1,640 ha (Gouvernement du Bénin, Nov 2023; AllAfrica, Aug 2026); Vallam Vadagal density of 2,313 beds per hectare (18,720 beds on 20 acres, India Infra Hub 2024); cost per bed US$2,100-4,500 (Exhibit 13). An illustration, not a forecast; bed ratios are scenarios.

Benin

GDIZ, 45 km north of Cotonou, had more than 25,000 jobs by August 2026, 16,000 of them in textiles, and aims for 300,000 by 2030. The government announced studies for worker housing in December 2023, after the first phase was under way. At the Tamil Nadu ratio, the 2030 target implies 141,000 beds on 61 to 122 hectares and about US$300m to 635m of capital. Benin's minimum wage of CFA 52,000, about US$89, covers a plain bed's capital charge at an assumed 6%, and the CFA franc, pegged to the euro, has held its dollar value since 2015 (Exhibit 20). The AfDB's US$100m investment in Arise IIP, which operates zones in Benin, Togo and Gabon, is the kind of platform finance to which a bed plan could be attached.

Nigeria

The Lekki Free Trade Zone covers about 155 km², and its first phase plans for 120,000 residents, according to the Czech embassy in Abuja. The Dangote refinery's contractor, ITB, reports partly building a camp for 22,000 construction workers and being awarded another for 10,000; in 2017 about 1,000 refinery staff had housing through third-party estates, according to Sweetcrude Reports. Camps planned for a second life could become the first operating beds. At the ₦70,000 minimum, about US$46 at the 2025 rate, a fifth of pay covers two thirds of a plain bed's capital charge at 6%, and the naira's fall from 192 per dollar in 2015 to 1,518 in 2025 makes local-currency finance essential.

Senegal and Côte d'Ivoire

Senegal's industrial sites agency APROSI reports 11,817 jobs and 71 companies across its sites, including the Diamniadio platforms, 30 km from Dakar. Its new city is planned for 350,000 residents in 45,000 homes, according to the agency developing it. As in Egypt, the new city is the housing answer, and a rental product for a worker's first years would complete it. In Côte d'Ivoire, IFC's local-currency loan to Addoha for 5,600 homes in Grand Abidjan shows the instrument a worker-bed facility could use.

141,000

beds GDIZ would need at its 300,000-job target at the Tamil Nadu ratio

4-7%

of GDIZ's 1,640 ha needed to hold them

32,000

construction bed-spaces in two camps reported at the Dangote refinery site

CHAPTER 7

07

The Heirstone bed test

Seven questions to ask of any new industrial park, or any phase of an existing one, before the first shed is let. Each draws on the evidence in this report.

The test is written for the people who decide what a park contains: zone developers and their masterplanners, the agencies that license zones, the manufacturers who commit to them and the lenders who finance both.

A park that can answer all seven has a housing plan. One that cannot answer the first three does not yet know whether it needs one.

The answers belong in the park's annual report. A bed ratio published every year, next to jobs and sheds, would give governments, buyers and lenders the measure that is missing today.

7

questions; the first three decide whether beds are needed, the last four how to deliver them

1 Where will the workforce come from?

Estimate the share of hires who will come from beyond daily commuting range. At Hawassa it was 91%; at Foxconn Sriperumbudur 80%. Where most hires are migrants, beds are part of the park's infrastructure, like power.

2 Can a bus reach the labour pool?

If a city with rental housing lies within reach, price the bus. Kenitra's free zone sits 10 km from the city and is served by 480 minibuses. Tangier paid about EUR 25 a worker a month in 2013, the same as the capital charge on a US$3,400 to 5,100 bed at an assumed 6 to 10%.

3 How many beds per 100 jobs, phase by phase?

Set a ratio for each phase and publish it next to jobs and sheds. Hawassa reached 7; Tamil Nadu's Vallam Vadagal provides 47 per 100 Foxconn jobs.

4 Does a fifth of the entry wage cover the bed's capital?

Compare 20% of the entry wage with the capital charge. Above about US$68 to 220 a month, depending on cost and assumed finance, rent covers the capital, though not land or running costs. Below it, name the payer: employer lease, grant or free land.

5 Will the room beat the private market?

Offer at least the space a worker already gets in a private room, 5.3 m^2 at Hawassa, and stay within the IFC and EBRD band of 4 to 5.5 m^2 and two to eight per room. Ashulia's 62-bed rooms stayed empty.

6 Who signs the lease and who runs the building?

Employers lease blocks and a professional operator runs them, as at Vallam Vadagal and as planned for the Hawassa dormitory. Beds let one by one by a government directorate, as at Ashulia, did not fill.

7 Is the land reserved in the masterplan?

China lets industrial projects raise the land share for offices and living facilities from 7% to 15%, mainly for dormitory rental housing. At Vallam Vadagal's density, 5% of Hawassa's land would house its entire workforce.

THE TEST APPLIED

Seven zones and two dormitories through the bed test

Exhibit 22 applies the seven questions to the zones with enough public data to try, and to the two dormitories in Chapter 4. The grey italic cells matter as much as the coloured ones: for most African zones the answers to the first questions are not published.

EXHIBIT 22

North African wages pass the rent test; beds per job, operators and data are the common gaps

The Heirstone bed test applied to selected zones and reference dormitories; latest public evidence

Meets the test Partly Fails

ZONE	1 WORKFORCE	2 REACH	3 BEDS PER 100 JOBS	4 RENT TEST	5 ROOM	6 LEASE AND OPERATOR	7 LAND
Hawassa, Ethiopia (2019-21)	91% from outside the city; beds for 8% of them	Walks of up to 3 hours a day	7; 27 if the dormitory is delivered	20% of base pay covers 38% of the capital charge; grant and leases fill the gap	Private rooms: three to a room, 5.3 m² each	Employer leases, private developer (planned)	Free land offered; dormitory at the park (planned)
Bole Lemi, Ethiopia (2018)	*Not published*	Inside Addis Ababa; private rentals nearby	None reported	Rent about 120% of base wage	*Not published*	*None reported*	*Not published*
Atlantic Free Zone, Morocco (2019)	*Not published*	10 km from Kenitra; 480 shift minibuses	None; buses instead	MAD 3,111 minimum: 20% covers any reference bed	*No housing*	Employers pay for buses	*Under 45 ha left in 2019*
10th of Ramadan, Egypt	*Not published*	49.3 km from Cairo; ~650,000 residents (NUCA)	55 homes (10 social) per 100 jobs	EGP 7,000: 20% covers a US$2,100 bed	*Family flats of 90 m²*	Homes sold to households	City-wide housing land
Tatu City, Kenya	*Not published*	*Not published*	23 residents; 5.5 affordable homes in the pipeline	*Not published*	*Not published*	Private city; homes for sale	5,000-acre mixed-use plan
GDIZ, Benin (2026)	*Not published*	45 km from Cotonou	None; study announced Dec 2023	CFA 52,000: 20% covers a US$2,100 bed at 6% only	*Not published*	*Not published*	*1,640 ha; housing land not published*
Sokhna, Egypt (2026)	*Not published*	*Not published*	*2018 plan; delivery not yet published*	EGP 7,000: 20% covers a US$2,100 bed	*Not published*	*Not published*	20 km² residential area in 2018 plan
INTERNATIONAL REFERENCES							
Vallam Vadagal, India (2024)	80% of Foxconn staff non-local; hostel built for them	Industrial-park housing for the plant	47; 92 with Foxconn's own blocks (reported)	*Rent not disclosed*	Six to a room	State-built; Foxconn runs it	20 acres for 18,720 beds
Ashulia, Bangladesh (2017)	*Not published*	*Not published*	744 beds; no tenants after a year	*Tk 1,000-1,200 a bed (US$12-15)*	62 to a room; 62 per toilet	Beds let one by one by a directorate	*Not published*

Source: Heirstone Worker Housing Benchmark 2026 and the sources for Exhibits 1 to 21. **Heirstone analysis.** Assessments are Heirstone judgements on public evidence; "not published" means we found no public source. The rent test compares 20% of the wage with the capital charge only, on assumed finance terms.

Two patterns stand out. First, the zones that fail on beds per 100 jobs are mostly those whose workers come from far away, or whose operators do not say. Hawassa's workforce was 91% migrant, and it had seven dedicated beds per 100 jobs.

Second, the North African zones pass the rent test that Ethiopia fails, and have housed workers through new towns and buses. Their next step is a rental product and an operator.

The test is cheap to run. Every input is something a park operator holds or could survey within months. Published yearly, the answers would let buyers and investors compare zones on a factor that, in Ethiopia, weighed on how many workers stayed.

RECOMMENDATIONS

Who should do what

Worker beds will not appear because a policy paper calls for them. They appear when an employer signs a lease, an operator runs the building, land is set aside and someone is named to pay any gap. Each audience controls one of those pieces.

Zone developers and operators

- Report beds per 100 jobs every year next to jobs, plants and sheds, by phase. None of the operators in our benchmark does.
- Reserve 5 to 10% of gross park land for worker rental housing in the masterplan. At Tamil Nadu densities, 5% of Hawassa would house its whole workforce.
- Offer dormitory blocks on the same terms as sheds: leased to employers, built to IFC and EBRD standards so that DFI-financed tenants can use them.
- Design construction camps for a second life as operating beds.

Manufacturers and anchor employers

- Put turnover in the housing business case. At 5 to 10% a month, a plant replaces its workforce every 10 to 20 months.
- Lease beds for the share of staff from out of town, and fold existing housing and transport allowances into the lease, as planned at Hawassa.
- Price the bus and the bed side by side: EUR 25 a month in Tangier would service the capital on a US$3,400 to 5,100 bed at an assumed 6 to 10%.
- Keep rooms at eight or fewer and above the private market's floor space; lenders applying IFC and EBRD guidance will check.

Governments and investment promotion agencies

- Egypt, Morocco, Tunisia: license rental dormitories on industrial land and let employers lease them; the wage covers a plain bed's capital charge.
- Ethiopia, Nigeria: budget the gap openly through employer leases, grants and free land, and publish occupancy.
- Kenya: let part of the housing levy fund employer-leased rental beds in SEZs.
- Benin, Senegal: fix a bed ratio for GDIZ and Diamniadio before the next phases; adopt a land rule like China's 15% allowance.

DFIs and lenders

- Open a worker-bed window: employer-leased rental blocks at the gate, financed in local currency as IFC did for Addoha in Côte d'Ivoire.
- Attach a bed plan to zone-platform finance, and treat the IFC and EBRD accommodation guidance as a design brief as well as a compliance check.
- Fund operators as well as buildings; the two dormitories in Chapter 4 suggest the operator decides take-up.

Housing developers and investors

- The product is a rental bed: four to six to a room, 4 to 5.5 m² a bed, US$2,100 to 4,500 of capital, leased to exporters.
- Start where wages clear the threshold, in Egypt, Morocco and Tunisia; blend in grants where they do not.
- Ask park operators for the bed ratio and the leasing commitments before committing capital.

How Heirstone works on industrial and worker real estate

We help landowners, zone developers, manufacturers, investors and governments decide what to build and what it is worth: market and demand studies, highest and best use and site selection, feasibility and financial models (DCF, IRR, NPV and Monte Carlo testing), masterplanning and phasing, and benchmarks of the kind in this report.

info@heirstoneconsulting.com
heirstoneconsulting.com

NOTES, SOURCES AND DEFINITIONS

Notes

Definitions

Dedicated worker housing: beds or homes provided for the workers of a park or employer, as opposed to general city housing. **Beds or homes per 100 jobs:** dedicated beds (or homes, where only homes are reported) divided by reported jobs, times 100; a home holds several people. **Entry wage:** the base wage of an entry-level worker or the statutory minimum wage, excluding allowances and bonuses. **Rent-to-wage:** monthly rent divided by the monthly entry wage. **Capital charge:** the monthly payment that repays the capital cost of a bed over the stated term at the stated rate. **Turnover:** workers leaving as a share of the workforce. **HFAE:** Housing for All Egyptians. **SEZ, EPZ, FTZ:** special economic, export processing and free trade zones.

Method

The Heirstone Worker Housing Benchmark covers 17 African zones and two international references, compiled from public sources current in 2026. Figures are as published, in the currency and period stated. Derived figures are labelled Heirstone analysis and use only the inputs shown. Capital charges use the capital recovery factor r/(1-(1+r)^-n), at 6% over 25 years (0.0782) and 10% over 20 years (0.1175); these terms are assumptions, not market quotations. They exclude land, operations, maintenance, utilities and vacancy, so true costs and subsidies are higher. Costs per bed from one country applied to another are reference benchmarks, not local estimates. Jobs, housing and wages often come from different years, and several currencies have devalued sharply over the period; ratios are indicative. Where sources conflict we show both or use the more conservative figure. Single-source and secondary figures are attributed in the text.

What we could not find

Completion and occupancy of the Hawassa worker dormitory begun in 2021; turnover data for any park outside Ethiopia; operating cost per bed for any project; worker housing or commuting data for Sokhna, Kigali, Diamniadio, Lekki or GDIZ; entry wages at Kenya's EPZs; rents at Vallam Vadagal; and the occupancy of Morocco's Chrafate homes. Each would sharpen the analysis, and most could be published by the operators concerned at little cost.

Use of this report

This report is published for general information. It is not investment, legal or valuation advice and should not be relied on as such. References to named projects and companies draw on public sources and imply no judgement of their conduct. Images are illustrative.

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SUMMARY AND CHAPTER 1

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CHAPTERS 2 AND 3

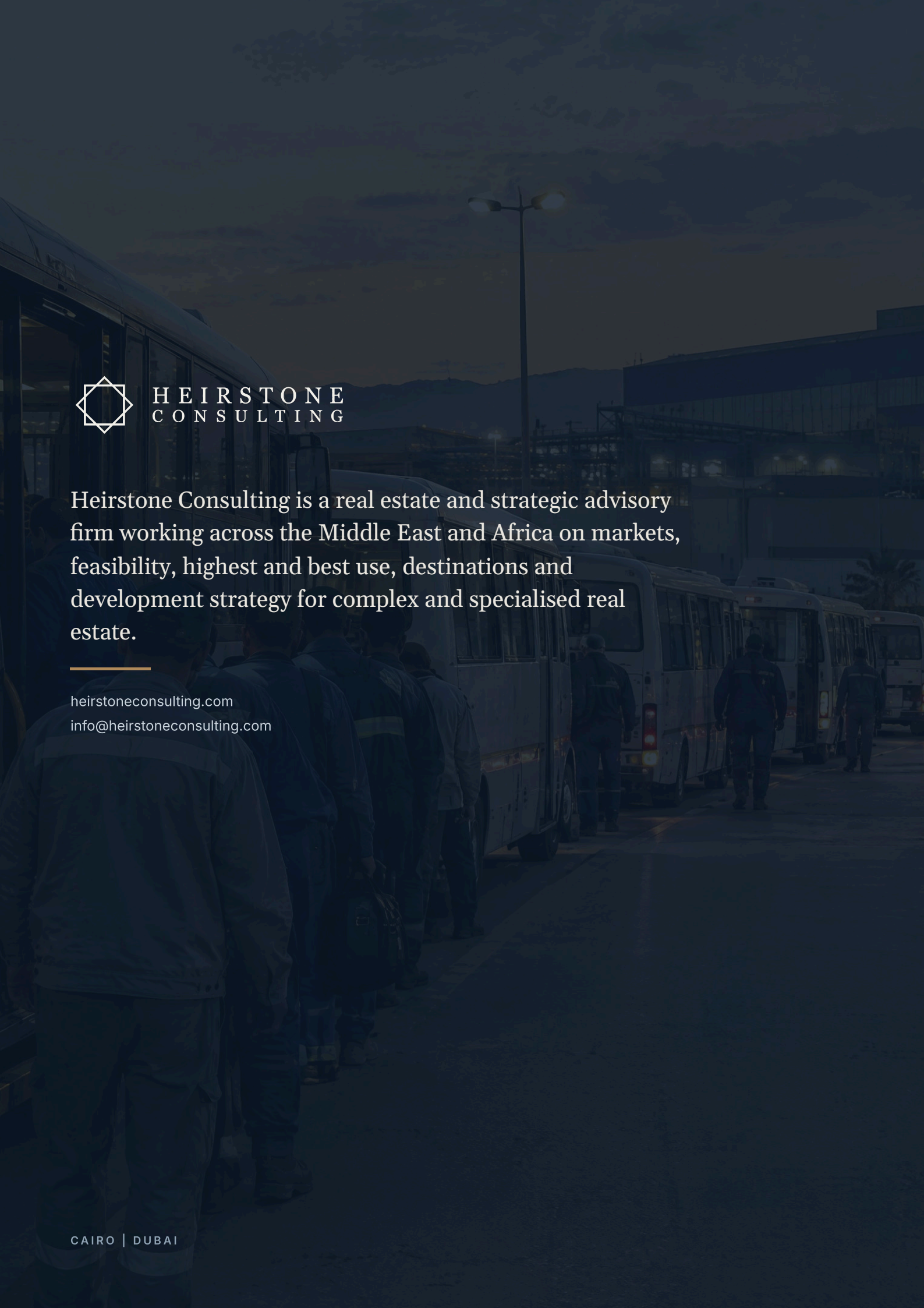
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heirstoneconsulting.com
info@heirstoneconsulting.com

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